



The Lux Institutional
Exchange:
A Non-Custodial, HFT-Grade
Venue for Crypto and
Tokenized Securities —
Nanosecond Matching,
Real-Time On-Chain Settle-
ment, Global Regulatory Reach

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Abstract

We describe the *Lux Institutional Exchange*: a white-label venue on which an operator runs a fully regulated market — Alternative Trading System (ATS), Broker-Dealer (BD), and Transfer Agent (TA) — whose trust-critical functions are *native protocol primitives of a sovereign Lux L1* rather than bolted-on vendors. The design targets institutional high-frequency trading and settles it with cryptographic finality. Its central properties are four. (i) **HFT-grade matching**: a deterministic central-limit-order-book engine matches in **169 ns** on average (p50 125 ns, p99 292 ns) at **5.91M orders/s** single-threaded and **11.88M orders/s** across ten threads, with 49 ns cancels and bit-exact CPU/GPU parity. (ii) **Non-custodial custody**: the venue never takes possession of client assets — balances sit under a dealerless threshold-MPC key that is never assembled at any single party, so the ATS is non-custodial by construction and carries none of the qualified-custodian single-point-of-compromise risk. (iii) **Real-time settlement**: fills settle atomically *inside the block*, delivery-versus-payment, at T+0 — the chain is the tape and the register. (iv) **Global regulatory reach**: jurisdiction is a declarative configuration axis, so the same core instantiates a U.S., Canadian, EU (MiCA), U.K., or Singapore venue by swapping a brand profile, with FHE-confidential compliance keeping client PII off-chain. We report the on-chain settlement primitive running live on Lux mainnet, and give the security and performance properties that make the architecture suitable for regulated, institutional trading of both crypto and tokenized securities.

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1 Introduction

An institutional trading venue is, structurally, four obligations bound together: it must *onboard* counterparties under KYC/AML rules (the broker-dealer function), *match* their orders fairly and at competitive speed and record the trades (the ATS function), *settle* and keep the authoritative ownership register (the transfer-agent function), and *custody* the assets throughout. Incumbents discharge each obligation with a separate vendor — a KYC provider, a matching engine, a books-and-records database, a qualified custodian — and the operator inherits the union of their risks. Two of those risks dominate. The *custodian* is a single institution (or HSM boundary) that,

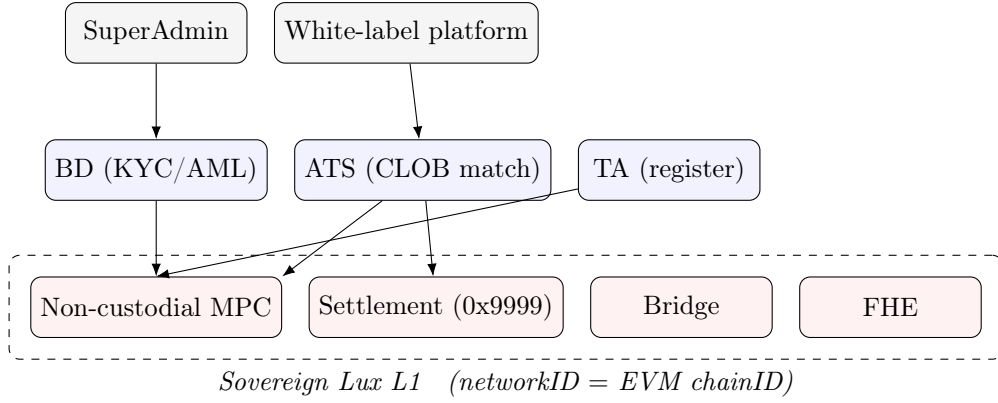


Figure 1: A role-scoped operator surface (ATS/BD/TA) and a white-label config surface over four native primitives of a per-tenant sovereign L1. Matching is HFT-grade; settlement and custody are on-chain and non-custodial.

if compromised or coerced, moves client assets unilaterally. And the *matching-plus-settlement* split forces a choice every venue makes badly: fast centralized matching with slow, trust-based settlement, or slow on-chain matching with no real throughput.

The Lux Institutional Exchange refuses both trade-offs. Matching runs at hardware speed in a deterministic order-book engine; settlement runs on-chain with cryptographic finality; and custody is *non-custodial* — assets never enter an exchange omnibus account, they move under a dealerless threshold key. The venue is delivered white-label: an operator supplies a declarative brand-and-jurisdiction profile and provisions a tenant. This paper specifies the architecture, grounds each claim in the shipped implementation and measured benchmarks, and states the properties that follow.

Contributions.

- An HFT-grade matching layer (§3) with published nanosecond-class latencies and multi-million-order-per-second throughput, with deterministic CPU/GPU parity.
- A *non-custodial* ATS custody model (§4) in which no party ever holds the signing key.
- Real-time, in-block, delivery-versus-payment settlement (§5), reported running live on Lux mainnet.
- A securities framework (§6) spanning crypto, tokenized public equities, and private offerings.
- Global regulatory reach as configuration (§7), with FHE-confidential compliance (§8).

2 Architecture

The system has three tiers (Figure 1): a *configuration* surface (SuperAdmin + tenant platform) that provisions and brands tenants and binds external providers; an *operator* surface (ATS/BD/TA) exposing the regulated functions as role-scoped consoles; and a *protocol substrate* of four native primitives — non-custodial matching-adjacent custody, on-chain settlement, bridging, and FHE confidentiality — running on a sovereign L1 per tenant, isolated at consensus (the tenant’s primary-network identifier equals its EVM chain identifier).

3 HFT-Grade Matching

The order-matching core is a deterministic central-limit-order-book (CLOB) engine. It is not a smart contract loop; it is a purpose-built matcher with a C++ hot path and a portable Go reference, benchmarked under `cmd/dexbench`.

Engine	Avg match	p50 / p99	Throughput
C++ (single thread)	169 ns	125 ns / 292 ns	5.91M orders/s
C++ (10 threads)	—	—	11.88M orders/s
Pure Go (<code>pkg/1x</code>)	381 ns	—	2.2M orders/s

Table 1: Measured matching performance (`luxfi/dex`). Order-cancel on the C++ engine is **49 ns**. Matching is deterministic per book, with bit-exact CPU/GPU parity, so a GPU-accelerated deployment reproduces the CPU result exactly rather than approximating it.

Property 3.1 (Nanosecond, deterministic matching). *The engine matches at a mean of 169 ns with a 292 ns 99th percentile and sustains millions of orders per second single-threaded. Determinism (per-book, CPU/GPU-identical) means the match sequence is reproducible and auditable — a requirement for a regulated ATS that a nondeterministic or approximate-accelerated engine cannot meet.*

Two architectural choices make this compatible with an on-chain venue rather than at odds with it. First, the **wire path is zero-copy**: the node’s ZAP transport removes serialization and HTTP/2 overhead on the hot path (it is several times faster to (de)serialize than protobuf and materially lower latency), so order ingress and market-data egress are not the bottleneck the matcher is fast enough to expose. Second, matching is *separated from settlement*: the engine produces fills at hardware speed, and those fills settle on-chain (§5) with cryptographic finality. The venue thus gets HFT latency *and* trustless settlement, instead of choosing one.

4 Non-Custodial Custody

Custody is where incumbent venues are weakest and where regulators scrutinize hardest: a qualified custodian, or the HSM inside it, is one entity that can move assets. The Lux venue removes that entity. Client assets are never held in an exchange account; they sit under a *dealerless threshold-MPC* key on the M-Chain.

Definition 4.1 (Dealerless threshold custody). *A group of n validators holds a t -of- n threshold signing capability such that (i) the private key is never computed or stored at any single party at any time, including at generation; (ii) any t parties jointly sign; and (iii) any coalition of fewer than t learns nothing about the key.*

Keys are generated by a verifiable secret-sharing DKG — each party samples its own polynomial and publishes commitments; the group key is the sum of contributions, so no dealer ever holds it. Signing is threshold ECDSA (CGGMP21-style) or Schnorr (FROST); the output is byte-identical to an ordinary signature and verifies under the standard verifier, so the scheme is transparent to every wallet and chain.

Property 4.1 (The ATS is non-custodial). *No single validator, no operator, and no platform ever holds material sufficient to move client assets. The exchange does not take custody; it coordinates a threshold that the client’s own key share participates in. Asset movement requires a t -of- n threshold, so there is no qualified-custodian single point of compromise and no exchange omnibus honeypot.*

For a regulated venue this is decisive: non-custodial operation reduces the operator’s custody liability and capital burden, and converts the strongest counterparty risk in the industry — “the exchange holds my assets” — into a cryptographic threshold the client participates in.

5 Real-Time On-Chain Settlement

Fills settle on a native settlement precompile at the reserved address `0x9999`. An order and its fill are ordinary transactions; the fill settles *atomically inside the block that includes it*, moving both legs — delivery of the asset and payment — in a single receipt. Settlement is therefore **real-time (T+0), delivery-versus-payment, and final on inclusion**: there is no T+1/T+2 window, no settlement-risk exposure between match and finality, and no separate clearing intermediary. The chain is both the trade tape and the ownership register, so the venue is self-auditing.

Live evidence. The settlement primitive is running on Lux mainnet. A maker bid and a taker fill executed against a live concentrated-liquidity pool: the taker sold 10 units of the base asset into a bid priced at 1.1, and the settlement receipt carried the on-chain fill event with output amount 11 together with both ERC-20 transfer legs in the same accepted transaction, conserving value exactly ($10 \times 1.1 = 11$). Real-time, atomic, on-chain settlement is a shipped, exercised primitive — not a specification.

6 The Securities Framework

One venue, one order book, one custody root, one register — across three asset classes:

- **Crypto** — spot markets (e.g. BTC, ETH, and native assets) under the same non-custodial custody and real-time settlement.
- **Tokenized public equities** — shares represented on-chain (identified by their real CUSIPs), traded on the ATS and registered by the TA, with corporate actions (dividends, splits) as native operations.
- **Private offerings** — Reg D 506(c) accredited raises, Reg A+ tiers, and Reg CF crowd-funding, issued through the TA and gated by FHE-evaluated accreditation checks, with an on-chain cap table.

Because the ATS/BD/TA roles are the same regardless of asset class, a single venue lists a crypto pair, a tokenized equity, and a private placement side by side — something no incumbent stack does without three separate systems.

7 Global Regulatory Reach

Each tenant runs its own sovereign primary network (network ID = EVM chain ID), isolated at consensus. A declarative brand profile then specializes it, and *jurisdiction is one of the axes*: the same substrate and the same operator services instantiate venues in different regulatory regimes by swapping providers and compliance parameters — not by rewriting code.

Remark 7.1. *The custody, matching, settlement, and confidentiality primitives are jurisdiction-invariant. Expanding to a new regime is a provider-binding and compliance-rules exercise, not a re-architecture — which is what makes a single white-label core viable as a global exchange platform.*

Jurisdiction	Regulatory mapping	Configured surface
United States	SEC / FINRA (ATS, BD, TA)	Reg D/A+/CF, OFAC, ACH rails
Canada	CIRO / provincial commissions	FINTRAC AML, Interac/EFT, CAD
European Union	MiCA / national NCAs	EU sanctions, SEPA rails, EUR
United Kingdom	FCA	UK sanctions, Faster Payments, GBP
Singapore	MAS	local KYC/AML, local rails

Table 2: Jurisdiction as configuration. The hard, trust-critical layer — matching, non-custodial custody, real-time settlement, confidentiality — is built and audited once; only the regulatory surface is localized per tenant.

8 FHE-Confidential Compliance

A chain-authoritative venue must not publish the client data that securities and privacy law protect. Sensitive state — beneficial ownership on the cap table, accreditation and sanctions attributes, pre-print order size — is stored as ciphertext, and the compliance predicates that gate activity (accreditation thresholds, sanctions/PEP screening, position and concentration limits) are evaluated *over the ciphertext* with lattice-based FHE, with threshold decryption gated by the MPC group.

Property 8.1 (Authoritative without disclosure). *For a compliance predicate P and encrypted client attributes $\text{Enc}(x)$, the venue computes $\text{Enc}(P(x))$ and reveals only the boolean outcome under threshold decryption; x is never disclosed to validators, the operator, or counterparties.*

9 Cross-Chain Bridging

Assets originating on other chains enter through the B-Chain bridge, attested by the *same* dealerless-MPC group that provides custody. A transfer is authorized by a threshold signature of the validator group; no bridge operator holds a unilateral minting key, and bridge fees are taken from the bridged amount. Because bridging shares the custody root, a bridged asset is held under the same non-custodial threshold guarantee as a natively issued one — there is no weaker bridge-custody tier.

10 Security and Performance Properties

1. **HFT matching.** 169 ns average match (p99 292 ns), up to 11.88M orders/s, deterministic with CPU/GPU parity (§3).
2. **Non-custodial.** No party holds the signing key; movement needs a t -of- n threshold; no omnibus honeypot (§4).
3. **Real-time settlement.** Atomic, in-block, DvP, T+0; finality on inclusion; live on mainnet (§5).
4. **Multi-asset.** Crypto, tokenized equities, and private offerings on one book and register (§6).
5. **Global.** Jurisdiction as configuration; one audited core, many regimes (§7).
6. **Confidential.** Compliance over ciphertext; client PII never on-chain in clear (§8).
7. **Tenant isolation.** Each tenant a sovereign L1 isolated at consensus.
8. **Post-quantum root.** Lattice-based FHE and the Lux post-quantum certificate/threshold constructions underpin confidentiality, finality, and custody.

11 Related Work and Conclusion

The design continues the “chain-first” line — the blockchain as authoritative source of truth for regulated financial infrastructure — and extends it from a settlement-and-registry substrate to a complete, institutional-grade venue. Its distinguishing choices are (i) an HFT matching core that makes on-chain settlement compatible with professional trading rather than a compromise against it; (ii) *non-custodial* custody, which removes the qualified-custodian single point of compromise instead of hardening it; (iii) real-time DvP settlement with cryptographic finality; and (iv) jurisdiction-as-configuration, which makes one audited core a global platform. We reported the settlement primitive running live on Lux mainnet and the matching benchmarks from the shipped engine. The result is a white-label substrate on which a regulated, high-frequency venue for crypto and tokenized securities — in any supported jurisdiction — is a configuration over a shared, cryptographically-grounded core.